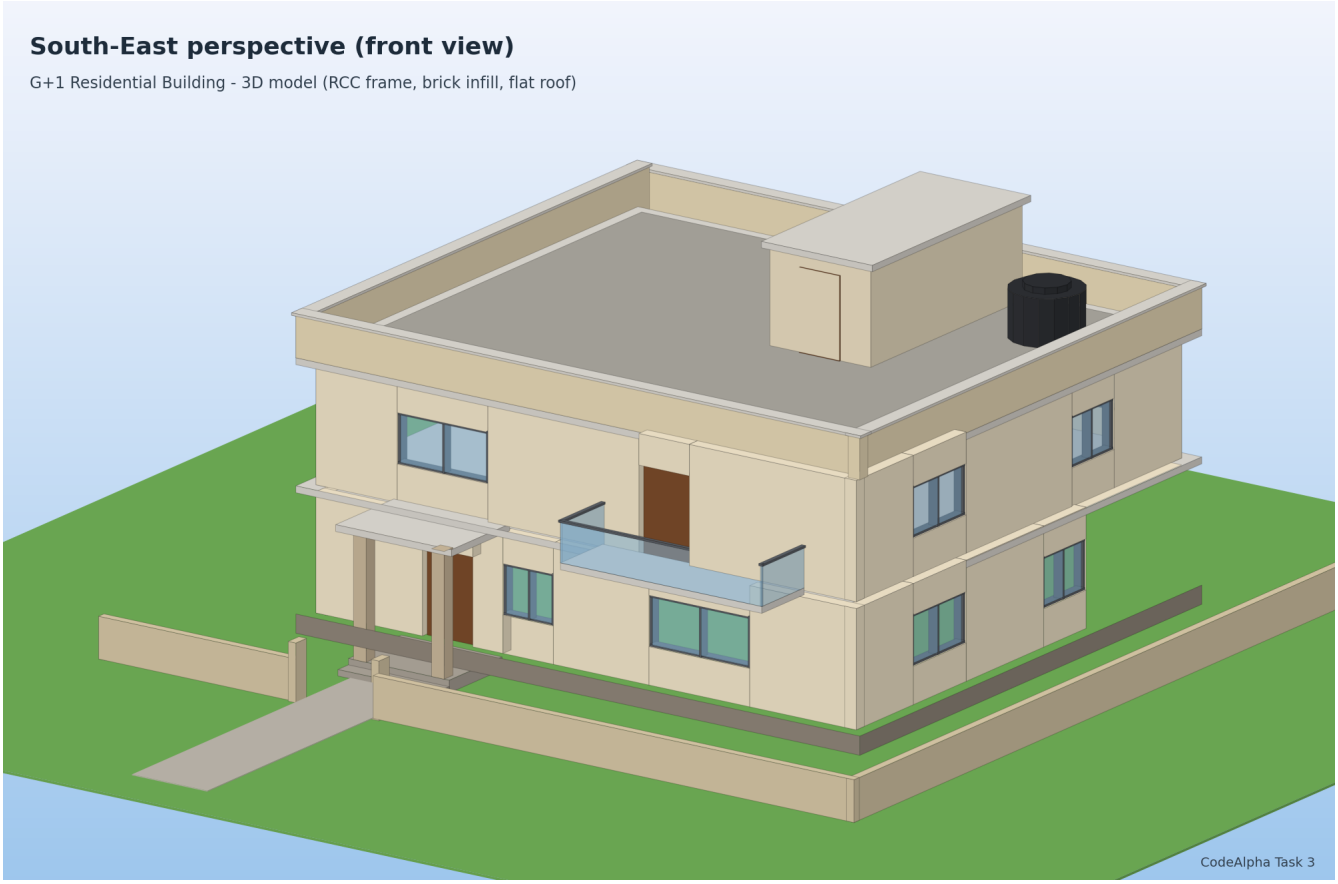
TASK 3 - 3D Modeling & Visualization

G+1 residential house | 3D DXF model + rendered images

1. Model description

A full 3D model of the G+1 house from Task 1 was built to the same dimensions as the 2D plan: RCC plinth, brick walls with real openings for every door and window, floor and roof slabs, parapet, staircase head-room (mumty), overhead water tank, entrance porch with canopy and columns, first-floor balcony with glass railing, compound wall with gate and a garden/path. The model is exported as a **3D DXF** (3DFACE entities on material layers) that can be opened directly in **AutoCAD** and imported into **SketchUp** or **Revit**; materials are applied by layer colour and transparency. The rendered images below are generated from the same geometry.

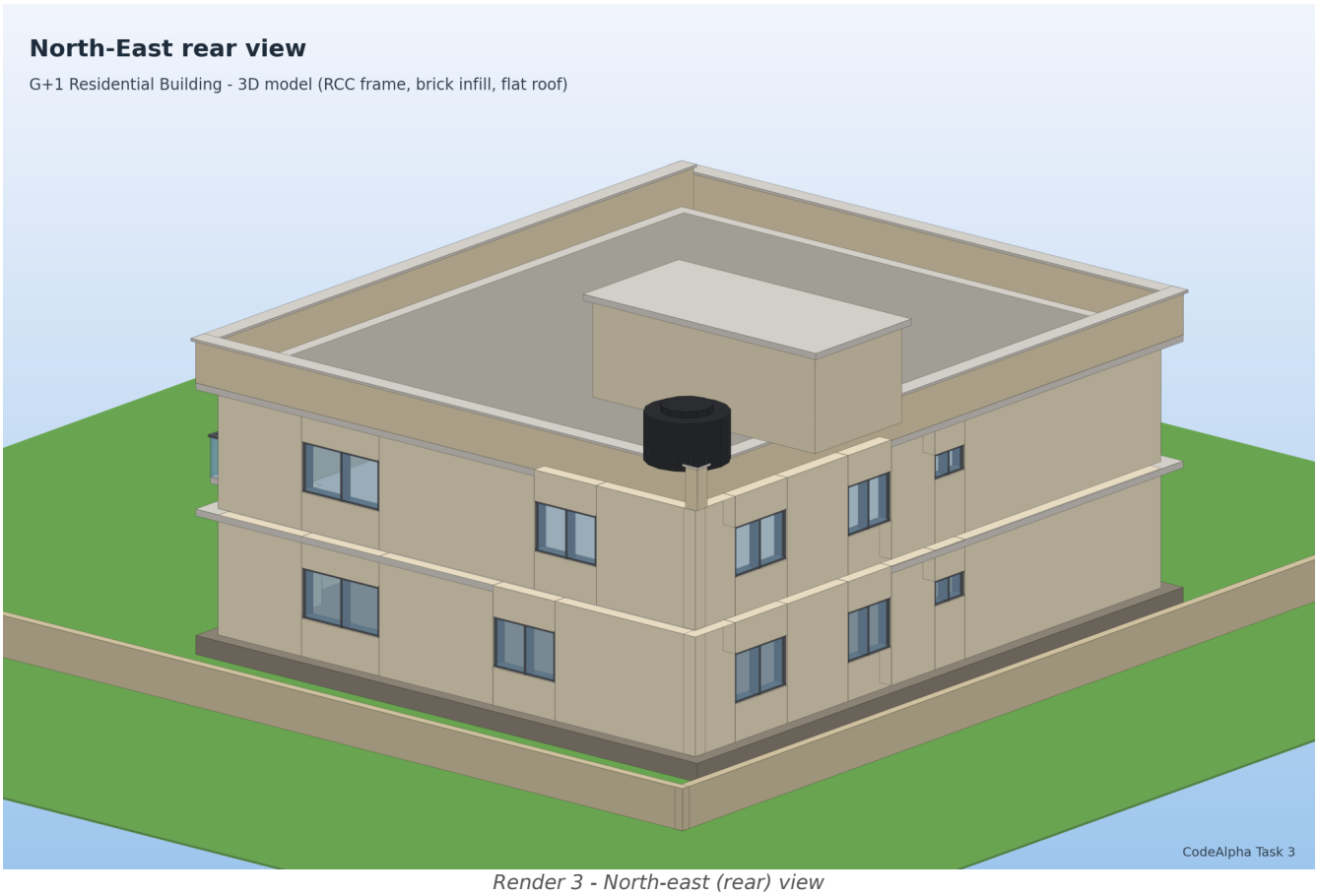
Component	Material / finish	Key dimension
Plinth	Stone-grey cement finish	Height 450 mm
External walls	Cream exterior plaster / paint	230 mm, floor height 3.0 m
Floor and roof slabs	Light-grey concrete, roof finish grey tiles	160 mm, overall height 6.45 m + 1.0 m parapet
Windows	Aluminium frames + blue-tinted 5 mm glass (55% transparent)	Sill 0.9 m, head 2.1 m
Doors	Teak-brown timber	2.1 m high
Balcony railing	Glass panel + steel handrail	1.0 m high, 4.0 x 1.2 m
Porch	RCC canopy on two pillars, three steps	2.0 x 1.5 m
Roof items	Dark-grey PVC water tank, stair mumty	Tank Ø 1.3 m, mumty 2.3 m high
Site	Lawn, paver path, 1.0 m boundary wall	Plot 15 m x 14 m



Render 1 - South-east perspective (main entrance, porch and balcony)



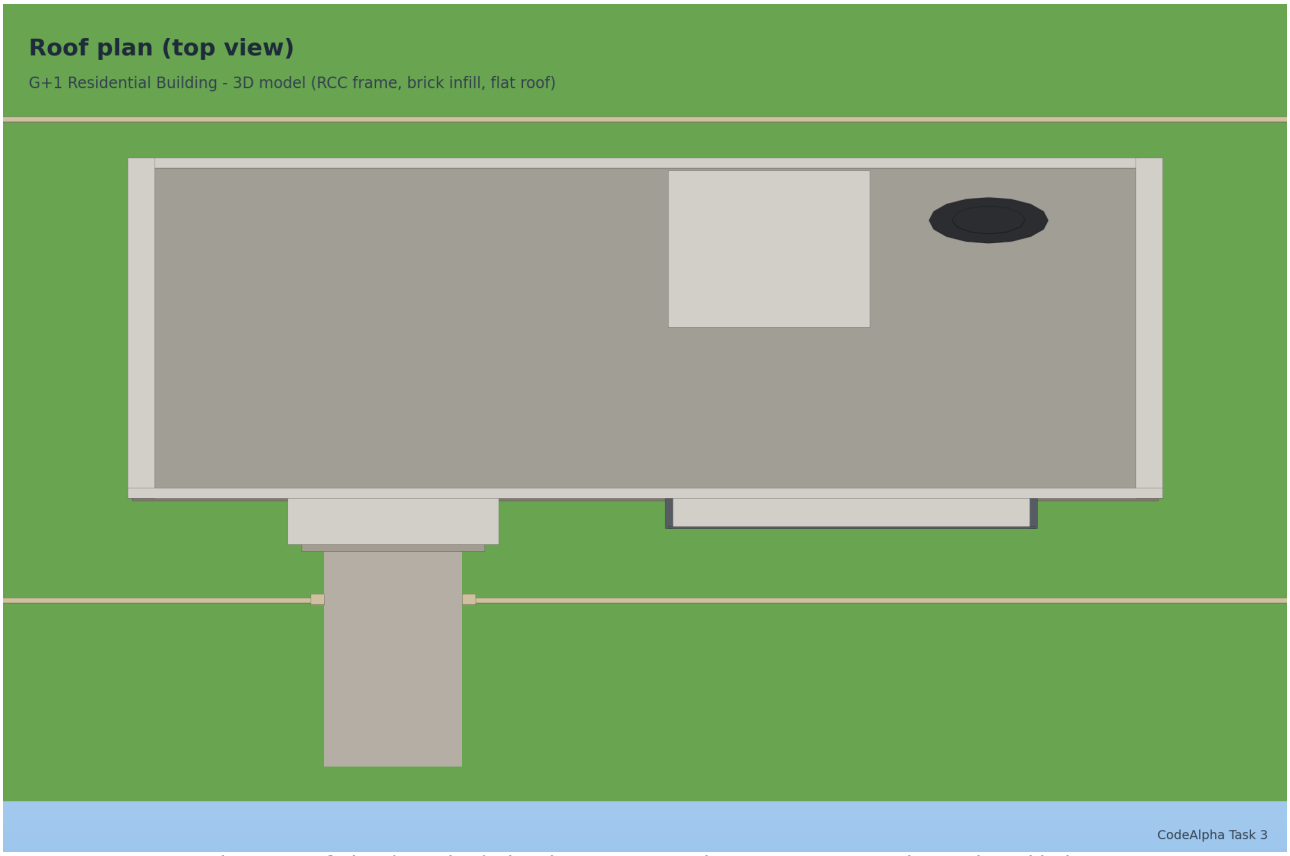
Render 2 - South-west perspective



Render 3 - North-east (rear) view



Render 4 - Front elevation (south)



Render 5 - Roof plan (top view) showing terrace, stair mumty, water tank, porch and balcony

Files submitted: CodeAlpha_Task3_GPlus1_3D_Model.dxf (3D design file) and the five rendered images (PNG) above. To reproduce a photo-realistic render in SketchUp/Revit, import the DXF, assign the named material layers (3D-WALL, 3D-GLASS, 3D-DOOR, ...) and render.